Seminar: Open-exam solutions

SOK-3081 | 2022 and 2019 open exams

How to use these solutions

These solutions match **Questions 1–16** and **BLANK1–BLANK34** in *S18_ch05_exam_seminar.qmd*. Exercise 1 retains the guided 2022 tuna example; Exercise 2 is the 2019 cross-sectional asparagus example.

Start each exercise at its own START UP in *S18_ch05_exam_seminar_solutions.R*. Each START UP clears the workspace and loads only that exercise's data, so Exercise 2 can be run independently. The solution uses *mosaic* and *car*, following the earlier lecture and exam scripts. Install these packages once if needed; an installation command is included as a comment in the R script. The code excerpts here do not execute when the document is rendered.

The solutions distinguish intended exam answers from ambiguous wording. Question 8 retains the 2022 normal-approximation option. Questions 13 and 14 use corrected 2019 answer lists. Price-index units are explained where the original exam uses the word percent.

Exercise 1: 2022 open exam

START UP

Start with a clean workspace, load the packages and download only the tuna data.

```
rm(list = ls())
options(scipen = 10, digits = 4)

library(mosaic)
library(car)

load(url("https://www.principlesofeconometrics.com/poe5/data/rdata/tuna.rdata"))
```

Question 1

The fitted equation is

$$\widehat{sal1} = 10531.30 - 25214.00\,apr1 + 8019.80\,apr2 + 9311.14\,apr3 + 11924.73\,dispad.$$

There are **52 observations**, **5 coefficients** including the intercept, and **47 residual degrees of freedom**. The price variables are in dollars; the dependent variable is cans sold per week.

R code

```
tuna_model <- lm(sal1 ~ apr1 + apr2 + apr3 + dispad, data = tuna)
summary(tuna_model)
```

Question 2

Field	Answer
BLANK1	one dollar
BLANK2	decrease
BLANK3	25214
BLANK4	one dollar
BLANK5	increase
BLANK6	8020
BLANK7	presence
BLANK8	11925

Completed interpretation: A **one-dollar** increase in brand 1's price is associated with a **decrease of 25214 cans** per week. A **one-dollar** increase in brand 2's price is associated with an **increase of 8020 cans** of brand 1 per week. With *dispad = 1*, fitted sales are **11925 cans higher**. Hold all other included variables constant.

Marking note: These are the intended exam answers. Because the question also offers "one cent", the combinations **one cent / decrease / 252** and **one cent / increase / 80** are also valid after rounding to the nearest can. The original options do not identify a unique unit-and-number combination.

R code

```
round(coef(tuna_model)[2])
round(coef(tuna_model)[3])
round(coef(tuna_model)[5])

round(coef(tuna_model)[2] * 0.01)
round(coef(tuna_model)[3] * 0.01)
```

Question 3

Correct option: A — 0.779.

The unrounded R^2 is approximately **0.7785**. The regressors explain about **77.9% of the sample variation in weekly cans sold**. This is a measure of fit, not evidence by itself of causation.

R code

```
round(summary(tuna_model)$r.squared, 3)
```

Question 4

Correct option: A — (-34768.04, -15659.97).

The 90% interval is

$$b_{apr1} \pm t_{0.95,47}\, se(b_{apr1}).$$

Holding the other regressors constant, a one-dollar price increase is associated with an estimated reduction in expected weekly sales of between about **15660 and 34768 cans**.

R code

```
round(confint(tuna_model, level = 0.90), 2)
```

Question 5

Field	Answer
BLANK9	-1.25
BLANK10	-4.79
BLANK11	16138
BLANK12	4214
BLANK13	inelastic
BLANK14	makes sense
BLANK15	higher
BLANK16	more

At the specified prices, fitted sales are **16138.30** cans with *dispad* and **4213.57** without it. The elasticity is

$$\hat{\varepsilon}_{apr1} = b_{apr1} \frac{0.80}{\widehat{sal1}},$$

giving **-1.2499** with *dispad* and **-4.7872** without it.

Wording note: The exam's intended phrase "more inelastic" means **less elastic in absolute value**. Both estimates still have absolute values greater than one, so demand is elastic in both scenarios. The own-price slope is the same; the fitted sales denominator changes.

R code

```
f <- makeFun(tuna_model)

f(0.8, mean(tuna$apr2), mean(tuna$apr3), 1)
f(0.8, mean(tuna$apr2), mean(tuna$apr3), 0)

round(coef(tuna_model)[2] * 0.8 /
        f(0.8, mean(tuna$apr2), mean(tuna$apr3), 1), 2)

round(coef(tuna_model)[2] * 0.8 /
        f(0.8, mean(tuna$apr2), mean(tuna$apr3), 0), 2)
```

Question 6

Field	Answer
BLANK17	16138
BLANK18	decreases
BLANK19	1733
BLANK20	10.74
BLANK21	drop

The baseline prediction is **16138.30 cans**. After the competitors' price cuts, the prediction is **14405.20 cans**.

$$\widehat{\Delta sal1} = -0.10\, b_{apr2} - 0.10\, b_{apr3} = -1733.09.$$

The signed percentage change is approximately **-10.74%**. In the exam sentence, "decreases by 1733" expresses the same result.

R code

```
before <- f(0.8, mean(tuna$apr2), mean(tuna$apr3), 1)
after <- f(0.8, mean(tuna$apr2) - 0.1, mean(tuna$apr3) - 0.1, 1)

before
after
after - before
100 * (after - before) / before
```

Question 7

Correct option: B — 710.7.

For coefficient order (intercept, *apr1*, *apr2*, *apr3*, *dispad*), define

$$c = (0, 0, -0.10, -0.10, 0)'.$$

Then

$$se(\widehat{\Delta sal1}) = \sqrt{c'\widehat{\mathrm{Var}}(b)c} = 710.7004.$$

Equivalently, the variance is $0.01\{\widehat{\mathrm{Var}}(b_{apr2}) + \widehat{\mathrm{Var}}(b_{apr3}) + 2\widehat{\mathrm{Cov}}(b_{apr2}, b_{apr3})\}$. The covariance term must be included. As in the earlier exam solution, the code below uses *deltaMethod()* to calculate the estimate and its standard error directly, including the covariance.

R code

```
change <- deltaMethod(
  tuna_model, "-0.1*b3-0.1*b4",
  parameterNames = paste("b", 1:5, sep = "")
)
change$Estimate
change$SE
```

Question 8

Original exam answer: E — (-3126.0, -340.15).

This option uses the **normal approximation**:

$$-1733.094 \pm 1.959964(710.7004) = (-3126.04, -340.15).$$

Chapter 5 t interval: Under the classical normal-error assumptions, use $t_{0.975,47}$ instead. The result is **(-3162.84, -303.35)**. None of the original options matches this t interval. Accept the correctly calculated t interval even though it has no matching letter.

Both intervals exclude zero and indicate lower expected sales after the competitor price cuts.

R code

```
deltaMethod(
  tuna_model, "-0.1*b3-0.1*b4",
  parameterNames = paste("b", 1:5, sep = ""),
  level = 0.95
)

tc <- qt(0.975, df.residual(tuna_model))
LB <- change$Estimate - tc * change$SE
UB <- change$Estimate + tc * change$SE
round(c(LB, UB), 2)
```

Exercise 2: 2019 open exam

START UP

Start a clean part: the previous exercise's objects are removed. Reload the packages and download only the asparagus data. Nothing from Exercise 1 is needed.

```
rm(list = ls())
options(scipen = 10, digits = 4)

library(mosaic)
library(car)

asparagus <- read.table(
  "https://userpages.umbc.edu/~coates/ec611/Waugh.txt",
  header = TRUE, nrows = 200
)
names(asparagus) <- tolower(names(asparagus))
```

Question 9

The fitted multiple regression is

$$\widehat{price} = 40.7613 + 0.137598\, green - 1.357256\, nostalks - 0.345283\, disperse.$$

There are **200 observations**, **4 coefficients** including the intercept and **196 residual degrees of freedom**. The dependent variable is the relative-price index $100P/PM$, not price in dollars. This is a cross-sectional comparison of asparagus bunches; there are no lagged variables.

R code

```
asparagus_model <- lm(price ~ green + nostalks + disperse, data = asparagus)
summary(asparagus_model)
```

Question 10

Field	Answer
BLANK22	two inches
BLANK23	increase
BLANK24	27.52
BLANK25	5
BLANK26	increase
BLANK27	6.79
BLANK28	3.45

Green: Two inches corresponds to **200 units** of *green*. Holding other characteristics constant, the fitted relative-price index increases by $200(0.137598) = 27.52$ points.

Number of stalks: Five fewer stalks gives a change of $-5(-1.357256) = +6.79$ index points.

Dispersion: An increase of **10 percentage points** in *disperse* gives $10(-0.345283) = -3.45$ index points, a reduction of **3.45**.

Units: The original exam calls these changes "percent". They are changes in the index $100P/PM$: one index point is 1% of the fixed benchmark PM. They are not percentage changes relative to a bunch's initial price.

R code

```
round(coef(asparagus_model)[2] * 200, 2)
round(coef(asparagus_model)[3] * -5, 2)
round(coef(asparagus_model)[4] * 10, 2)
```

Question 11

Correct option: A — (-8.27, -5.30).

For five additional stalks, the estimated index change is **-6.7863** and its standard error is **0.7541**.

$$5b_{nostalks} \pm t_{0.975,196}\, 5\, se(b_{nostalks}) = (-8.2735, -5.2991).$$

Both the estimate and its standard error must be multiplied by five. Option C gives the interval for just one additional stalk.

R code

```
change <- deltaMethod(
  asparagus_model, "5*b3",
  parameterNames = paste("b", 1:4, sep = "")
)
change$Estimate
change$SE

round(5 * confint(asparagus_model, level = 0.95)[3, ], 2)
```

Question 12

Field	Answer
BLANK29	95.23
BLANK30	drops
BLANK31	77.97
BLANK32	17.26

At the sample means of *green* and *nostalks*, the fitted relative-price indices are **95.2311** when *disperse = 0* and **77.9669** when *disperse = 50*.

$$\widehat{\Delta} = 50 b_{disperse} = -17.2641.$$

The original fill-in answer **17.26** describes the magnitude of the drop. The signed change, new minus original, is **-17.26 index points**. Relative to the initial price, the percentage change is about **-18.13%**, not -17.26%.

R code

```
f <- makeFun(asparagus_model)

before <- f(
  green = mean(asparagus$green),
  nostalks = mean(asparagus$nostalks), disperse = 0
)
after <- f(
  green = mean(asparagus$green),
  nostalks = mean(asparagus$nostalks), disperse = 50
)

before
after
after - before
100 * (after - before) / before
```

Question 13

Corrected option: D — 6.48.

With *green* and *nostalks* unchanged, the difference is $50 b_{disperse}$. Therefore,

$$se(\widehat{\Delta}) = 50\, se(b_{disperse}) = 50(0.1296563) = 6.4828.$$

The original exam's options did not contain this standard error. The updated worksheet corrects the option list. The standard error of a difference is not obtained by adding the standard errors of the two predicted levels.

R code

```
change <- deltaMethod(
  asparagus_model, "50*b4",
  parameterNames = paste("b", 1:4, sep = "")
)
change$Estimate
change$SE
```

Question 14

Corrected option: C — (-34.13, -0.40).

Using the t distribution with **196 residual degrees of freedom**,

$$\widehat{\Delta} \pm t_{0.995,196}\, se(\widehat{\Delta}) = -17.2641 \pm t_{0.995,196}(6.4828) = (-34.1269, -0.4014).$$

The interval excludes zero, indicating a lower expected relative-price index at *disperse = 50* than at *disperse = 0*, holding the other characteristics constant.

The original exam's answer list consisted of individual numbers rather than intervals. The worksheet supplies corrected intervals and explicitly requests the t distribution. The normal approximation **(-33.96, -0.57)** is a distractor.

R code

```
tc <- qt(0.995, df.residual(asparagus_model))
LB <- change$Estimate - tc * change$SE
UB <- change$Estimate + tc * change$SE
round(c(LB, UB), 2)
```

Question 15

Correct option: A — (84.46, 53.76, 115.16).

For *green* = *550*, *nostalks* = *20* and *disperse* = *14*, the fitted index is **84.4612**. Its **95% prediction interval** is **(53.7638, 115.1586)**.

This predicts the relative-price index of **an individual bunch** with these characteristics. A confidence interval for the conditional mean would be narrower because it excludes the additional individual-observation error variance.

R code

```
f(green = 550, nostalks = 20, disperse = 14, interval = "prediction")

f(green = 550, nostalks = 20, disperse = 14, interval = "confidence")
```

Question 16

Field	Answer
BLANK33	decreases
BLANK34	38

Since $price = 100P/PM$,

$$P = \frac{PM}{100}\, price.$$

For ten additional stalks,

$$\widehat{\Delta P} = \frac{2.782}{100}(10 b_{nostalks}) = -0.37759 \text{ dollars} = -37.7589 \text{ cents}.$$

The price therefore **decreases by about 38 cents**, or **0.38 dollars**, holding the other characteristics constant. The original question's opening sentence asks for dollars, while the fill-in text asks for cents; both units are given here.

R code

```
round(coef(asparagus_model)[3] * 10 * 2.782 / 100, 2)

round(coef(asparagus_model)[3] * 10 * 2.782)
```